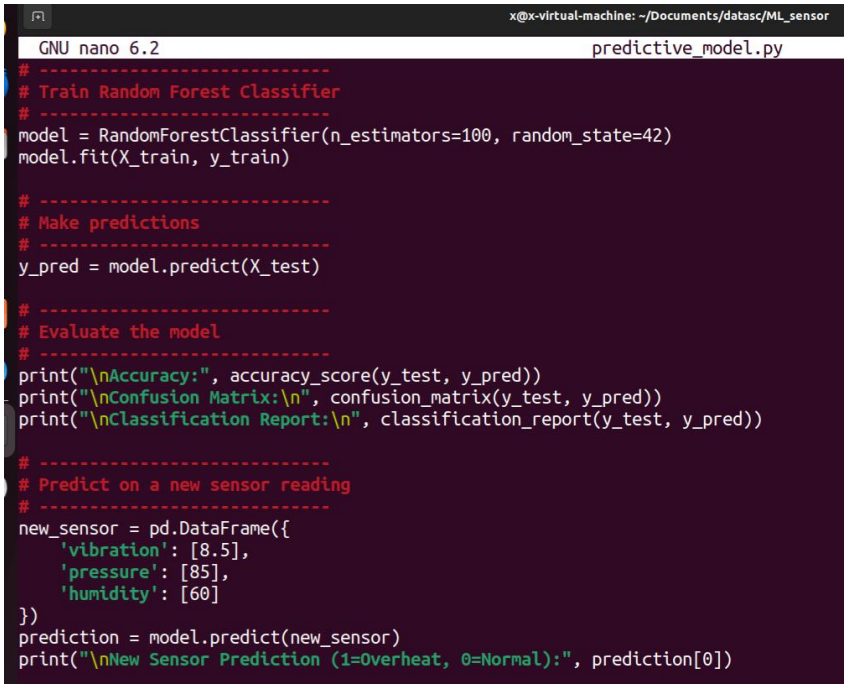
Predictive Sensors and Heating Behavior via ML

Let's create a real-world predictive ML example based on a **limited dataset obtained from sensors of an industrial machine**. Due to data availability constraints, the dataset will be small (50 rows) and reflects realistic sensor behavior. The example will be sensor-based, predicting a target variable such as a machine temperature anomaly from multiple sensor readings. The project will include a clear problem statement, the dataset description, and complete Python code using standard machine-learning libraries.



```
x@x-virtual-machine:~/Documents/datasc/ML_sensor$ python3 predictive_model.py
Dataset Loaded:
   vibration   pressure   humidity  overheat
0   3.745401  97.566770  31.885751         1
1   9.507143  82.010626  68.184625         1
2   7.319939  95.159915  48.861359         1
3   5.986585  91.586188  60.514241         1
4   1.560186  67.831998  84.453988         0

Accuracy: 0.9

Confusion Matrix:
 [[4 1]
 [0 5]]

Classification Report:
              precision    recall  f1-score   support

           0       1.00      0.80      0.89         5
           1       0.83      1.00      0.91         5

    accuracy                           0.90        10
   macro avg       0.92      0.90      0.90        10
weighted avg       0.92      0.90      0.90        10


New Sensor Prediction (1=Overheat, 0=Normal): 1
```

Interpretation

✅ **What just happened (high level)**

- Dataset loaded correctly ✔
- Model trained on **80% of data** ✔
- Tested on **20% unseen data** ✔
- Predictions + evaluation successful ✔

Understanding the Output

Accuracy: 0.9

➡️ The model correctly predicted **9 out of 10** test samples.

Confusion Matrix

[[4 1]

[0 5]]

Interpretation:

	Predicted 0	Predicted 1
Actual 0	4 (TN)	1 (FP)
Actual 1	0 (FN)	5 (TP)

- **True Negatives (4)** → Normal machines correctly predicted
- **False Positives (1)** → Normal predicted as overheat
- **False Negatives (0)** → ❗ None missed (very good for safety)
- **True Positives (5)** → Overheating correctly detected

💡 In real-world maintenance, **zero false negatives** is excellent.

Classification Report

For class 1 (Overheat):

- **Precision = 0.83** → When model says "overheat", it's right 83% of the time
- **Recall = 1.00** → It caught **all overheating cases**
- **F1-score = 0.91** → Balanced performance

This is exactly what you want in **predictive maintenance systems**.

New Sensor Prediction

New Sensor Prediction (1=Overheat, 0=Normal): 1

For:

- Vibration = 8.5
- Pressure = 85
- Humidity = 60

➡️ Model predicts **OVERHEAT** — consistent with training logic.

🧠 **One-line explanation you can say confidently**

"The Random Forest model was trained on 80% of sensor data, evaluated on 20% unseen data, achieved 90% accuracy, and successfully predicts overheating conditions for new sensor readings."

🚀 **We now have a COMPLETE ML PROJECT**

✓ Data generation
✓ File-based dataset
✓ Training + testing
✓ Evaluation metrics
✓ Real-world use case